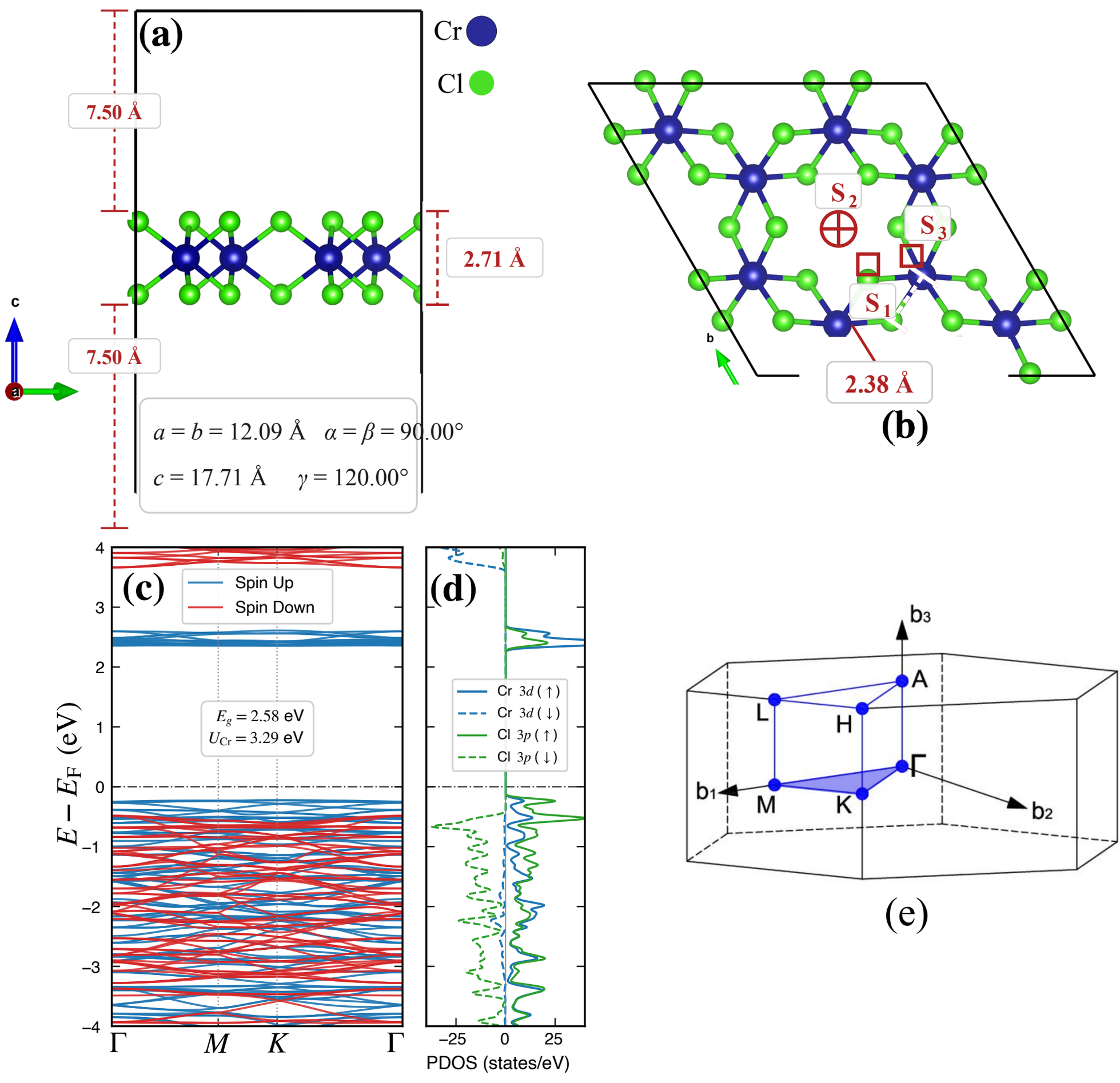

(a)
Cr
Cl
7.50 Å
2.71 Å
7.50 Å
c
a
$a = b = 12.09$ Å $\alpha = \beta = 90.00°$
$c = 17.71$ Å $\gamma = 120.00°$
S_2
S_3
S_1
b
2.38 Å
(b)
(c)
Spin Up
Spin Down
$E_g = 2.58$ eV
$U_{Cr} = 3.29$ eV
$E - E_F$ (eV)
Γ
M
K
Γ
(d)
Cr 3d (↑)
Cr 3d (↓)
Cl 3p (↑)
Cl 3p (↓)
PDOS (states/eV)
b3
A
L
H
Γ
b1
M
K
b2
(e)